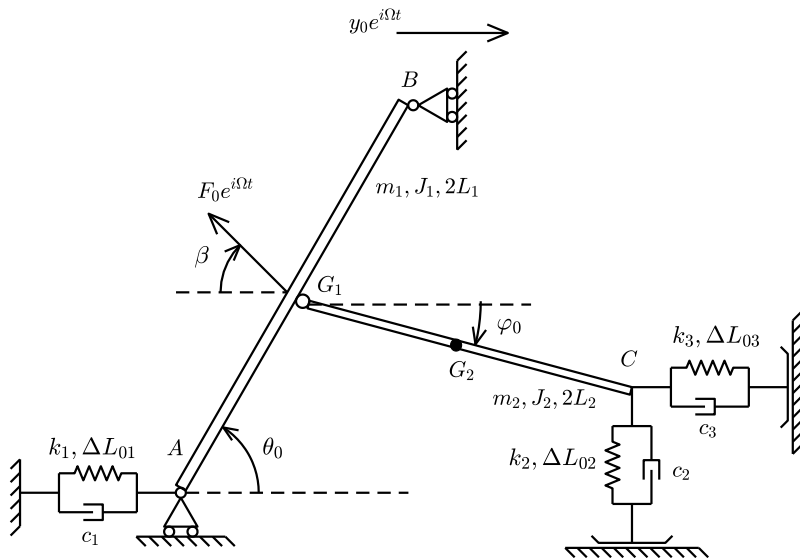


Dynamics of Mechanical Systems

January 30th, 2020 – standard problem (grading up to 30/30)



$m_1 = 6 \text{ kg}$	$k_1 = 5000 \text{ N/m}$
$m_2 = 10 \text{ kg}$	$k_2 = 5000 \text{ N/m}$
$J_1 = 0.08 \text{ kgm}^2$	$k_3 = 5000 \text{ N/m}$
$J_2 = 0.3 \text{ kgm}^2$	$c_1 = 2 \text{ Ns/m}$
$L_1 = 0.2 \text{ m}$	$c_2 = 2 \text{ Ns/m}$
$L_2 = 0.3 \text{ m}$	$c_3 = 2 \text{ Ns/m}$
$\Delta L_{01} = 0.02 \text{ m}$	$\theta_0 = \pi/3$
$\Delta L_{02} = 0.0356 \text{ m}$	$\varphi_0 = \pi/6$
$\Delta L_{03} = 0.0787 \text{ m}$	$\beta = \pi/4$
$y_{02} = 0.02 \text{ m}$	$F_{01} = 100 \text{ N}$
$\varphi_{02} = \pi/6$	$\varphi_{01} = \pi/3$
$\Omega_2 = 14\pi$	$\Omega_1 = 10\pi$

The mechanical system above lays in the vertical plane and is shown in its equilibrium position.

The students are requested to:

- Copy the table shown below on top of the first page of each sheet and write their surname, name, identification number and SNxxx code (see note below), **together with the columns titles** (Surname, Name...), leaving blank the field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;

Surname	Name	Matriculation number	SNxxx code	Nr.

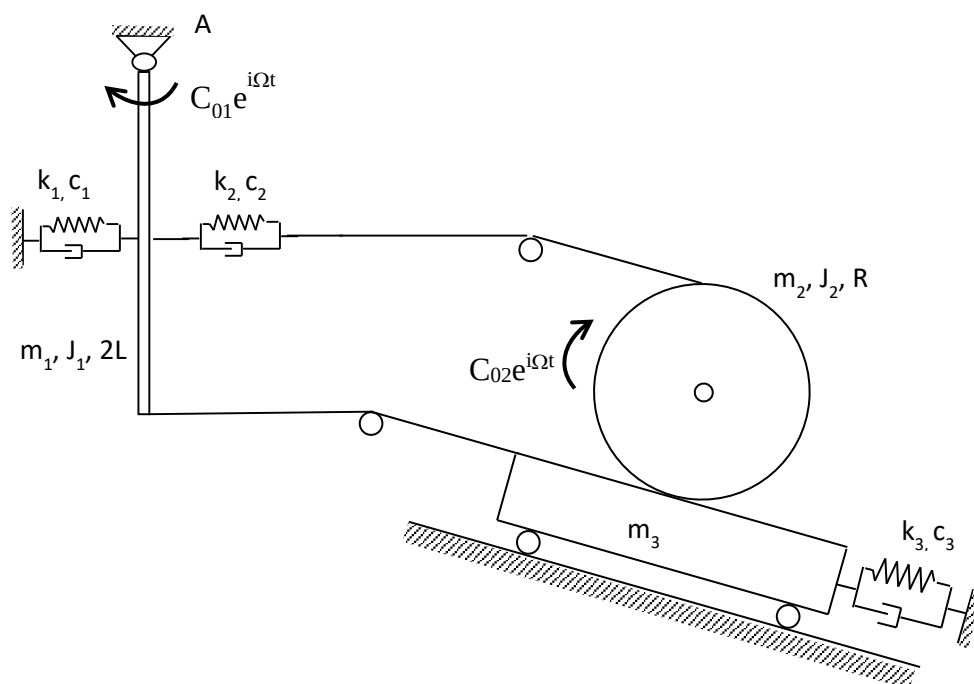
- Write the linearized equations of motion around the given equilibrium configuration;
- Compute the natural frequencies and modes of vibrations;
- Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0e^{i\Omega t}$. Output: the horizontal component of the displacement of point G_1 and the vertical component of the displacement of point G_2 . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
- Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0e^{i\Omega t}$. Output: the elastic and damping force transmitted by the spring/damper system (k_3, c_3). Save the graphical results as SNxxx3.fig and SNxxx4.fig
- Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0e^{i\Omega t}$. Output: the dynamic component of the constraint forces at carts A and B. Save the graphical result as SNxxx5.fig
- Compute the time history (time step = 0.001 s, $T = 2$ s) applying the harmonic constraint displacement $y_{02}e^{i(\Omega_2 t + \varphi_2)}$ and the harmonic horizontal force $F_{01}e^{i(\Omega_1 t + \varphi_1)}$. Output: of the constraint force applied on cart A. Save the graphical result as SNxxx7.fig;
- Compute the static spring preloads ΔL_{02} and ΔL_{03} when $\theta_0 = \pi/4$ and $\varphi_0 = \pi/3$ keeping the same value of ΔL_{01} .

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

- Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
- Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
- Select the folder *Test January 30th 2020 - standard*
- Click on button Add and select Single document;
- Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
- Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
- Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems

January 30th, 2020 – simplified problem (grading up to 22/30)



The mechanical system in the picture lays in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding along the rigid body m_3 . The students are requested to:

- Copy the table shown below on top of the first page of each sheet and write their surname, name, identification number and SNxxx code (see note below), **together with the columns titles** (Surname, Name...), leaving blank the field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;

Surname	Name	Matriculation number	SNxxx code	Nr.

- Write the linearized equations of motion around the given static configuration;
- Compute the natural frequencies and modes of vibrations;
- Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_{01}e^{i\Omega t}$ at the vertical bar. Output: the rotation of the disk and the rotation of the vertical bar. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
- Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_{02}e^{i\Omega t}$ at the disk. Output: the absolute displacement of the disk's center along the inclined plane and the horizontal constraint force in the hinge A. Save the graphical results as SNxxx3.fig and SNxxx4.fig.

$m_1 = 10 \text{ kg}$	$k_1 = 1000 \text{ N/m}$	$c_1 = 1 \text{ Ns/m}$
$m_2 = 5 \text{ kg}$	$k_2 = 2000 \text{ N/m}$	$c_2 = 2 \text{ Ns/m}$
$m_3 = 10 \text{ kg}$	$k_3 = 1000 \text{ N/m}$	$c_3 = 1 \text{ Ns/m}$
$J_1 = 0.2 \text{ kgm}^2$	$L = 0.5 \text{ m}$	
$J_2 = 0.5 \text{ kgm}^2$	$R = 0.25 \text{ m}$	

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

- Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
- Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
- Select the folder *Test January 30th 2020 - simplified*
- Click on button Add and select Single document;
- Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
- Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
- Confirm the procedure clicking the button Publish.